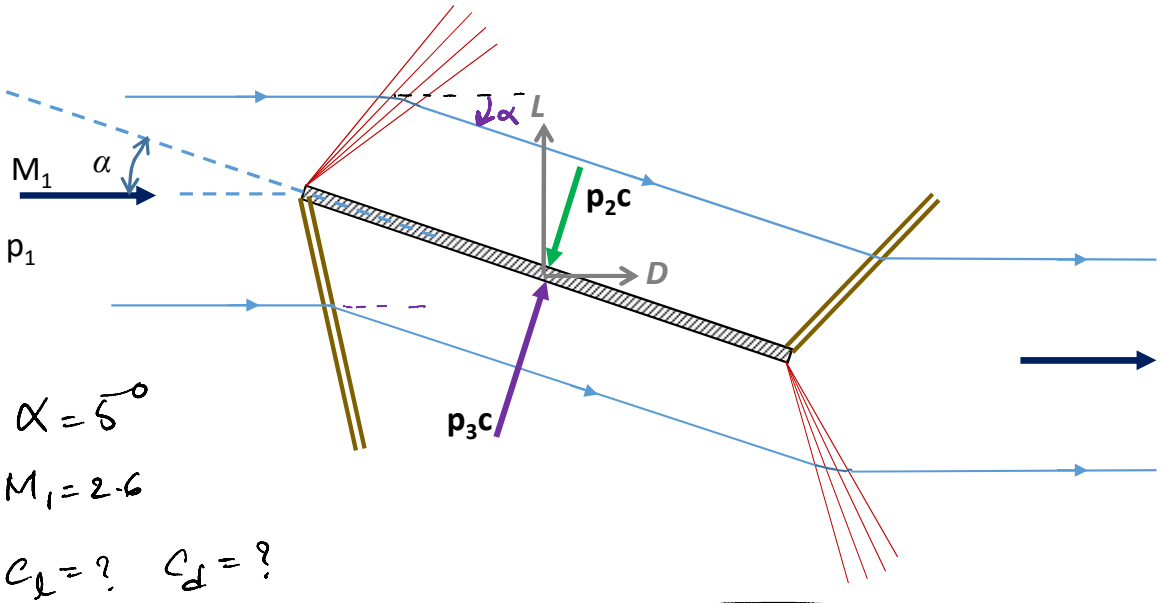
Numericals on Shock-Expansion Theory

Problem:

A thin flat plate is kept at an angle of attack of 5 deg. In a flow of Mach 2.6. Calculate the Lift and Drag coefficients.



$\alpha = 5^\circ$

$M_1 = 2.6$

$c_l = ?$ $c_d = ?$

Lift per unit span

$$L = (p_3 - p_2)\, c \cos\alpha$$

Drag per unit span

$$D = (p_3 - p_2)\, c \sin\alpha$$

Lift Coefficient

$$c_l = \frac{L}{\frac{1}{2}\rho_\infty V_\infty^2 c} = \frac{L}{q_\infty c}$$

Dynamic pressure $q_\infty = \frac{1}{2}\rho_\infty V_\infty^2$

$$= \frac{1}{2}\frac{\gamma p_\infty}{\gamma p_\infty}\rho_\infty V_\infty^2 = \frac{1}{2}\gamma p_\infty \frac{\rho_\infty}{\gamma p_\infty} V_\infty^2$$

(we know $\sqrt{\frac{\gamma p_\infty}{\rho_\infty}} = a_\infty$)

$$= \frac{1}{2}\gamma p_\infty \frac{V_\infty^2}{a_\infty^2} = \frac{\gamma}{2} p_\infty M_\infty^2$$

$$q_\infty = \frac{\gamma}{2} p_\infty M_\infty^2$$

$$C_l = \frac{L}{q_1 c} = \frac{(p_3 - p_2) c \cos\alpha}{\frac{\gamma}{2} p_1 M_1^2 c} = \frac{2}{\gamma M_1^2}\left(\frac{p_3}{p_1} - \frac{p_2}{p_1}\right)\cos\alpha$$

$$\Rightarrow \quad C_l = \frac{2}{\gamma M_1^2}\left(\frac{p_3}{p_1} - \frac{p_2}{p_1}\right)\cos\alpha$$

Drag Coefficient

$$C_d = \frac{D}{q_1 c} = \frac{(p_3 - p_2) c \sin\alpha}{\frac{\gamma}{2} p_1 M_1^2 c} \Rightarrow C_d = \frac{2}{\gamma M_1^2}\left(\frac{p_3}{p_1} - \frac{p_2}{p_1}\right)\sin\alpha$$

$\frac{p_2}{p_1}$: 1-2: Expansion fan

- Flow deflection angle $\theta_2 = \alpha = 5^\circ$

For $M_1 = 2.6$ $\xrightarrow{\text{from P-M tables}}$ $\nu(M_1) = 41.42$

$$\theta_2 = \nu(M_2) - \nu(M_1)$$

$$\Rightarrow \quad \nu(M_2) = \theta_2 + \nu(M_1)$$

$$= 5 + 41.42 = 46.42$$

For $\nu(M_2) = 46.42$ $\xrightarrow{\text{From P-M tables}}$ $M_2 = 2.84$

For $M_1 = 2.6$ $\xrightarrow{\text{Isentropic tables}}$ $\frac{p_1}{p_{01}} = 0.0501$

For $M_2 = 2.84$ $\xrightarrow{\text{Isentropic tables}}$ $\frac{p_2}{p_{02}} = 0.0347$

$$\frac{p_2}{p_1} = \frac{p_2}{p_{02}} \times \frac{p_{02}}{p_{01}} \times \frac{p_{01}}{p_1}$$

(Across an expansion wave $p_{01} = p_{02}$)
(Isentropic)

$$= 0.0304 \times 1 \times \frac{1}{0.0501}$$

$$\frac{p_2}{p_1} = 0.693$$

$\frac{p_3}{p_1}$: 1-3 : oblique shock

- flow deflection angle $\theta_1 = \alpha = 5^\circ$

$M_1 = 2.6$ and $\theta_1 = 5^\circ$ $\xrightarrow{\theta\text{-}\beta\text{-}M \text{ chart}}$ $\beta_1 = 26.5^\circ$

$$M_{n_1} = M_1 \sin\beta_1 = 2.6 \sin 26.5^\circ = 1.16$$

For $M_{n_1} = 1.16$ $\xrightarrow{\text{Normal shock tables}}$ $\frac{p_3}{p_1} = 1.403$

$$C_l = \frac{2}{\gamma M_1^2}\left(\frac{p_3}{p_1} - \frac{p_2}{p_1}\right)\cos\alpha = \frac{2}{1.4 \times 2.6^2} \times \left(1.403 - 0.693\right) \times \cos 5^\circ$$

$$C_l = 0.15$$

$$C_d = \frac{2}{\gamma M_1^2}\left(\frac{p_3}{p_1} - \frac{p_2}{p_1}\right)\sin\alpha =$$

$$C_d = 0.0131$$

Home work:

Problem:
A thin flat plate is kept at an angle of attack of 15 deg. in a flow of Mach 2.6. Calculate the Lift and Drag coefficients.
Solution: lift coefficient=0.45 and Drag coefficient=0.12

Problem:
A thin flat plate is kept at an angle of attack of 30 deg. in a flow of Mach 2.6. Calculate the Lift and Drag coefficients.
Solution: lift coefficient=1.19 and Drag coefficient=0.69